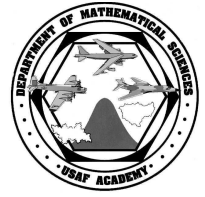




DEPARTMENT OF THE AIR FORCE

THE DEPARTMENT OF MATHEMATICAL SCIENCES

USAF ACADEMY, COLORADO



8 August 2025

MEMORANDUM FOR ALL CADETS ENROLLED IN DATA 421/422

FROM: HQ USAFA/DFMS, Maj Justin Merrick, 333-2124

SUBJECT: Class of 2025 Data Science Capstone Course Letter

1. **Introduction.** Welcome to the Data Science Capstone Experience. The purpose and goal of this two course sequence is for data science majors to:
 - i. apply techniques learned through their course of study,
 - ii. demonstrate the ability to acquire new data-science methodologies and tools,
 - iii. develop their independent research abilities,
 - iv. enhance their technical communication skills, and
 - v. refine their problem-solving abilities within the framework of an applied research project.All students will be required to meet certain expectations and requirements over the course of this academic year. This document seeks to make those expectations and requirements clear.
2. **Selecting a Capstone Project.** A curated list of capstone projects will be provided by the course director, Maj Merrick. Students will form groups of three to four students and rank-order preferences. Based on these preferences and other commitments made to USAF capstone partners, cadets will be assigned to projects by the course director.

Cadets who fail to submit preferences or a proposal will be randomly assigned to a research group by the course director. A failure to submit preferences will adversely affect final course grades.
3. **Capstone Mentor and Customer.** Your capstone project will be conducted under the guidance of a technical mentor. Projects cultivated by Maj Merrick have a mentor pre-assigned and, in most cases, include a USAF (or external) customer.
4. **Non-curated Capstones.** Necessary modifications will be made for cadets absent from USAFA (e.g., those studying abroad) based on their mission requirements at their assigned duty location. If this applies to you and you have not already done so, please reach out ASAP to Maj Merrick. If you have other extenuating circumstances, you may discuss the option of pursuing a non-curated capstone individually with Maj Merrick. Note that the course director has received guidance that there is a high bar for students desiring to complete an individual capstone. Additional details in this regard are included in the “Data Science Capstone Instructions” document.
5. **Required Deliverables.** Each group must submit the following deliverables to receive credit for both Data 421 (Fall) and Data 422 (Spring). The required submission format (group or individual) is specified in parentheses. **Please note that all dates listed in this document are subject to change. If any dates do change, cadets will be notified accordingly.**

Data 421 – Fall

- **Routine Check-ins with Capstone Mentor (Group).** Your team must maintain regular contact with your capstone mentor. The nature and frequency of this communication should be agreed upon with your mentor. For instance, a weekly meeting or email may be sufficient. **Weekly communication is advised.** The course director does not need to be included in this

correspondence. However, the course director will consult with mentors when assigning grades to evaluate how well the group adhered to the agreed schedule.

Bi-weekly Progress Reports to CD (Group). This bi-weekly progress report must document the work completed since the last submission, the planned activities for the next reporting period, observations, and any encountered problems. Students are encouraged to discuss topics they found interesting and why they found them interesting. Finally, please provide specifics regarding each group member's efforts; individual projects may obviously forgo such detailed accounting. The form of this report is left to each group to develop. Possible structures include a professionally formatted e-mail, a memorandum or bullet background paper formatted according to AFH 33-337, The Tongue and Quill, or a Markdown file created from meeting notes from the last sprint retrospective. Progress reports should be sent to justin.merrick@afacademy.af.edu. A due date for the initial progress report will be set with Maj Merrick after groups and projects have been assigned but must be NLT T8 (28 August).

If your report indicates insufficient effort is being made, the course director may call you in for further discussion. While we expect progress in projects, our primary concern is your commitment to learning and improving as a data scientist. **This means you should document your failures as well.**

- **Problem Framing and Literature Brief (Group).** Present an introduction to the problem and a literature review to the class on lessons T15 and T16 (18 and 22 September). If needed, T14 (16 September) will be reserved as an overflow day. This brief should introduce the problem, relevant concepts, and review a few academic papers that address similar issues. You may also need to explain new topics to the class, such as the difference between explainable and interpretable AI. The brief may also include discussion topics to brainstorm with the class and inform the group's future endeavors.

In preparation for this presentation, each group must schedule a mandatory 20-minute pre-brief vector check with Maj Merrick in his office NLT T14 (16 September). Please ensure your group is prepared for the pre-brief with a draft slide deck. It's acceptable if the deck is partially complete, but include placeholders to outline the flow of your presentation.

- **Update and Discussion Brief 1 (Group).** Scheduled for December 1 and 3, with December 5 reserved as an overflow day if needed. This presentation aims to provide a semester-end status update to the class. Cover current progress, including explanations of methodologies explored (for example, if using an unfamiliar approach, explain how it works), exploration of models created, and/or a discussion of preliminary results. Embrace failed models and results as they can lead to insightful discussions.
- **Team Assessment (Individual).** Provide a self-assessment of your team's work and recommend a letter grade for your contributions between 1 and 5 December, but NLT COB on 5 December 25. Additionally, assess each team member's contribution. Send this assessment to justin.merrick@afacademy.af.edu.
- **Draft Final Report (Group).** Provide a formal first draft of your final report that includes (at least) an introduction and literature review. Due NLT COB on lesson 40 (5 December).

Data 422 – Spring

- **Bi-weekly Progress Reports to CD (Group).** This bi-weekly progress report must document the work completed since the last submission, the planned activities for the next reporting period, observations, and any encountered problems. Students are encouraged to discuss topics they

found interesting and why they found them interesting. Finally, please provide specifics regarding each group member's efforts; individual projects may obviously forgo such detailed accounting. The form of this report is left to each group to develop. Possible structures include a professionally formatted e-mail, a memorandum or bullet background paper formatted according to AFH 33-337, The Tongue and Quill, or a Markdown file created from meeting notes from the last sprint retrospective. Progress reports should be sent to justin.merrick@afacademy.af.edu. The due dates for these progress reports are 2359 on the lessons/dates below:

Lesson	Date
T2*	9 January
M7	23 January
M12	6 February
T16	20 February
T26**	20 March
T31	10 April

* Use this opportunity to plan out the semester

** Before leaving for Spring Break

- **Update and Discussion Brief 2 (Group).** Scheduled for T17-T20 (24 Feb, 26 Feb, 2 Mar, 4 Mar). Presentations on T20 will be by cadets doing research in other departments ONLY. This presentation aims to provide a mid-semester status update to the class. Utilizing feedback from the previous Update and Discussion Brief, groups shall: cover current progress, including explanations of methodologies explored, exploration of models created, and a discussion of preliminary results. Embrace failed models and results as they can lead to insightful discussions. This brief shall be a stand-alone briefing.
- **Final Outbrief (Group).** Scheduled for T36-40 (30 Apr, 4 May, 6 May, 8 May). Presentations on T36 will be by cadets doing research in other departments ONLY. This final brief should be provided in a manner that fully explains the problem, explains your methodology and rationale behind its development, and discusses your results.
- **Final Report (Group).** A final report must be submitted with the following six sections:
 - i. an abstract,
 - ii. an introduction,
 - iii. a literature review,
 - iv. a methodology section,
 - v. a results section, and
 - vi. a conclusion/recommendation section.

Reports will be graded based on the quality of their communication. Please be cogent, accurate, and formal in your prose. Final reports will be due NLT COB on lesson T40 (8 May 26).

- **Team Assessment (Individual).** Provide a self-assessment of your team's work and recommend a letter grade for your contributions between 4 and 8 May, but NLT COB on 8 May 26. Additionally, assess each team member's contribution. Send this assessment to justin.merrick@afacademy.af.edu.
6. **Grades.** Grades at Prog and the end of semester will be assigned holistically based upon all deliverables and reports provided by the cadets. The holistic standards are based upon the five course goals presented previously. Namely, the following holistic summaries characterize each letter grade.
- **A:** Cadets effectively utilized tools learned during their course of study and explored novel data science methodologies and concepts, integrating them into their work. Independent research was of high quality, demonstrating thorough investigation and innovative thinking. Professionalism

was evident in all interactions: cadets maintained a high standard of conduct when communicating with sponsors, peers, and faculty, demonstrating respect, reliability, and initiative. Cadets consistently maintained open, transparent, and timely communication with their sponsors, addressing questions and concerns proactively and seeking feedback regularly. Briefings were well-prepared, clear, and comprehensive, effectively summarizing progress, challenges, and next steps in a concise and professional manner. Communication with the sponsor and within the team was always clear, respectful, and goal-oriented, ensuring effective collaboration and timely updates. Verbal and written communication was cogent, accurate, and formal, demonstrating clarity and professionalism in all deliverables and reports.

- **B:** Cadets utilized tools learned during their course of study and explored novel data science methodologies and concepts but could have further developed the integration of these tools. Independent research was solid but could be more comprehensive. Professionalism was maintained in most interactions, but there were occasional lapses in timeliness or thoroughness when engaging with sponsors, peers, or faculty. Cadets maintained open communication with the sponsor but might have missed opportunities for more frequent or proactive updates. Briefings were generally well-prepared but could have been more concise or focused on the most critical aspects of the project. Communication was generally clear and respectful, though some interactions could have been more proactive or detailed. Verbal and written communication was mostly accurate and formal, with some areas needing improvement in clarity, professionalism, or thoroughness.
- **C:** Cadets utilized tools learned during their course of study but did not explore novel data science methodologies or concepts. Independent research was limited or not fully developed, and there was a lack of depth in problem-solving. Professionalism was inconsistent: while cadets generally maintained a respectful attitude, their engagement with sponsors and team members lacked the initiative and reliability expected. Cadets communicated with the sponsor, but responses were often delayed or lacked sufficient detail. Briefings were underdeveloped or lacked focus, missing important context or relevant details. Communication with the sponsor, team, or faculty was sometimes unclear or unprofessional, with limited updates provided. Verbal and written communication showed some gaps in professionalism, clarity, and formality, with room for improvement in terms of conciseness and relevance.
- **D:** Cadets utilized few tools learned during their course of study and did not explore novel data science methodologies or concepts. Independent research was minimal or incomplete, and there was little initiative taken in resolving problems or seeking new solutions. Professionalism was lacking: cadets frequently missed deadlines, provided incomplete updates to sponsors, and showed a lack of collaboration within the team. Communication with the sponsor was infrequent, late, or lacked necessary detail, and briefings were poorly prepared or incomplete. Cadets were often unresponsive or unclear in their interactions with the sponsor and team, hindering project progress. Verbal and written communication was often lacking professionalism, coherence, and clarity, with significant room for improvement in both formal language and team coordination.
- **F:** Cadets incorrectly utilized tools learned during their course of study, failing to explore novel data science methodologies or concepts. Independent research was either nonexistent or of very poor quality. Professionalism was severely lacking: cadets were frequently unresponsive or disrespectful in communications with the sponsor, team, or faculty, missed deadlines without explanation, and failed to contribute meaningfully to the team's efforts. Communication with the sponsor was poor or non-existent, and briefings were absent or entirely ineffective, lacking essential context or actionable insights. Cadets demonstrated no initiative to maintain open communication or follow-up with the sponsor, negatively impacting the project. Verbal and written communication was sloppy, unclear, and unprofessional, with little attention to detail or appropriate tone.

Feedback will be provided on each of these deliverables to the students in relation to these holistic standards. Data 421/422 will be graded separately using the above rubric based upon all deliverables provided in the considered class. Your capstone mentors will also be contacted regarding their assessment of your work and progress.

7. **Course Meetings.** Over the next two semesters, most of the time you will be working individually and consulting with your capstone mentor. However, sometimes we will meet as a group during our scheduled class period. There is a planned meeting schedule below, but you should be aware that this schedule is subject to change. Note that if the course director calls a meeting on a given lesson, you are expected to attend. Remember that attending class ranks very high on the cadet schedule order of precedence, if appointments are made during our class period, you may have to break them if a class meeting is called.
8. **Academic Standards.** You are expected to abide by the AF Cadet Wing Honor Code Reference Handbook (b), the Academics with Honor Policy (c), Section VI, Academic Policies and Procedures of the Dean of Faculty Handbook (a), and USAFA Instruction 36-3534, Violations of Academic Standards (d) and the DFMS Policy on Electronic Devices During Graded Events (e) at all times. Failure to adhere to these policies will result in an academic penalty and, if deemed appropriate, referral to the Cadet Honor System, but note that Academic penalties are independent of the Cadet Honor System. All policy letters are posted in the classroom and can be found on the course team on Teams. Failure to adhere to these policies could result in an academic penalty. **Never attempt to claim credit for someone else's work, including when that work was done by an AI tool.**
9. **Use of Artificial Intelligence/Large Language Model Tools:** You are encouraged to use Generative AI up to Level 5 – unrestricted, attributed use of GenAI – as defined in the DF Generative AI Usage Policy Memo, published 22 July 2025. Be aware that sponsors may restrict this usage further for their projects and/or it may be restricted by the classification of systems or data that you will be using. Be sure to accurately document your use of AI tools in any presentation you give and in your written reports. **Never attempt to claim credit for someone else's work, including when that work was done by an AI tool.**
10. **Dignity and Respect.** You are a valuable member of our Air Force & Space Force and deserve to be treated with dignity and respect. Likewise, you are expected to treat your classmates and instructor with dignity and respect. Our goal is to create an environment where everyone is valued and has an equal opportunity to be heard and contribute. Be thoughtful, considerate, aware, forgiving, kind, and willing to admit mistakes. Your instructor will do the same. If you have any concerns talk with your instructor, or go directly to the Department Head, Col Scott Williams, or his Deputy, Lt Col Christopher McGrath. There are also Ombuds in DF outside of the Mathematical Sciences Department that will address your concerns. Whatever avenue you choose, please know that Col Williams and I are committed to dignity and respect for all cadets and faculty.
11. **Contact Information.** Please contact me if you have any questions. My office is 6D-198 in Fairchild Hall. My email is justin.merrick@afacademy.af.edu and my office number is 333-2124. You can use Teams to chat with me. I'm looking forward to a great capstone experience!

//SIGNED//

JUSTIN MERRICK, Maj, USAF
Assistant Professor of Mathematical Sciences
Course Director, Data 421/422

Course Schedule.

Tentative Capstone class meetings Fall 2025		
Lesson	Date	Purpose
1	8 Aug	Admin, expectations for first semester
2 – 5	12 – 20 Aug	Special Topics
13	12 Sep	ACC operational insights
15 – 16	18 – 22 Sep	Problem Framing and Literature Brief (PF&L Brief)
38 – 40	1 – 5 Dec	Update and Discussion Brief 1 (Upd. & Disc. Brief 1)

Tentative Capstone class meetings Spring 2026		
Lesson	Date	Purpose
1	7 Jan	Data Cards & Data Versioning
2	9 Jan	xAI & Model Cards
3	13 Jan	MLOps
4	15 Jan	Storytelling
6	22 Jan	Special Topics Overflow
17 – 19	24 Feb – 2 Mar	Update and Discussion Brief 2 (U&D 2)
20	4 Mar	External Capstone Progress Briefs
37	30 Apr	External Capstone Final Briefs
38 – 40	4 – 8 May	Final Brief

Recall that the course director may call for additional class meetings as needed. In that case, all students are expected to attend during the regularly scheduled class meeting time.